

(TR-104) PYTHON WITH AI/ML

Training Day 24 Report

Date: 12 February 2026

Topic Covered: Spam Detection Models – Naive Bayes & Logistic Regression (NLP Classification)

The twenty-fourth day of training focused on implementing spam detection models using classical NLP techniques. The day involved applying machine learning algorithms on real datasets and understanding how text classification works in practice.

- Implemented Spam Detection using Naive Bayes on a publicly available dataset.
- Applied text preprocessing and vectorization techniques before model training.
- Trained and evaluated a Logistic Regression model for spam classification.
- Used TF-IDF representation for converting text into numerical features.
- Compared performance of Naive Bayes and Logistic Regression models.
- Evaluated models using accuracy, precision, recall, and F1-score.
- Understood model behavior on real-world dataset variations.
- Practiced complete NLP pipeline including preprocessing, training, and prediction.
- Analyzed overfitting and generalization in classification models.
- Strengthened understanding of evaluation metrics and train-test split importance.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/NLP/blob/main/Text_Classification/logistic_regression_spam_classifier.ipynb
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/spam-classifier-using-naive-bayes-and-logistic-regression-in-python-3b042dec86b3>